Training Day 4

1. Topics Covered and Brief Explanation

1. Introduction to Python

Python is a high-level, interpreted and general-purpose programming language known for its simple syntax and readability. It is widely used in Artificial Intelligence, Machine Learning, Data Science and automation.

2. Features of Python

Studied the major features of Python:

- Easy to learn and understand.
- Simple and readable syntax.
- Open-source programming language.
- Large collection of libraries.
- Supports object-oriented programming.
- Platform independent.

3. Variables in Python

Variables are used to store data values in a program. Python does not require explicit declaration of variable types.

Example:

```
name = "Neha"
age = 21
marks = 95.5
print(name)
print(age)
print(marks)
```

4. Data Types in Python

Learned about commonly used Python data types:

Data Type	Example
Integer	age = 21
Float	price = 99.5

String	name = "Neha"
Boolean	is_valid = True
List	numbers = [1, 2, 3]
Tuple	values = (10, 20)
Dictionary	student = {"age": 21}

5. Operators in Python

Studied different types of operators:

- Arithmetic operators: +, -, *, /, %
- Comparison operators: ==, !=, >, <
- Logical operators: and, or, not
- Assignment operators: =, +=, -=

6. Input and Output Operations

Learned how to accept input from users using input() and display results using print().

2. Tools and Technologies Used

Tool	**Purpose**
Python 3	Writing and executing Python programs.
Google Colab	Practising Python programming online.
Google Chrome	Accessing online Python learning resources.

3. Learning Outcomes

- Understood the fundamentals and features of Python programming.
- Learned how to declare variables and work with different data types.
- Gained knowledge of arithmetic, comparison and logical operators.
- Practised taking user input and displaying output.
- Developed a basic understanding of Python programming for future AI and ML applications.